

Atomic layer deposition of energy band tunable tin germanium oxide electron transport layer for the SnS-based solar cells with 400 mV open-circuit voltage ^{EP}

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ABSTRACT

Tin germanium oxide, (Sn,Ge)O₂, films were prepared using atomic layer deposition and tailored to a SnS absorber layer by incorporating various amounts of germanium into tin oxide to adjust band alignments at the interfaces of SnS/(Sn,Ge)O₂ photovoltaic devices. Carrier concentrations of (Sn,Ge)O₂ were suppressed from 10²⁰ to 10¹⁸ cm⁻³ with germanium incorporation, with nitrogen doping further reducing carrier concentrations by another order of magnitude. Excellent tunability of both band energy levels and carrier concentrations of (Sn,Ge)O₂ allowed optimizing SnS-based solar cells. SnS/(Sn,Ge)O₂:N devices were demonstrated, with an open-circuit voltage as high as 400 mV, due to the effective mitigation of interfacial recombination of photogenerated carriers at the SnS/(Sn,Ge)O₂:N absorber-buffer heterojunction interface.

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Tin monosulfide (SnS) is a promising earth abundant and non-toxic p-type semiconductor material for photovoltaic applications. SnS thin films have displayed many desirable properties including a suitable bandgap of 1.1 eV, high absorption coefficients exceeding 10¹⁴ cm⁻¹, and sufficiently high mobilities ($\mu \sim 10 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$), making it an attractive alternative to leading thin film absorber materials.¹⁻⁵ The better performing SnS-based devices utilize a tunable n-type buffer layer to optimize the conduction band offset (CBO) at the p-n junction interface. Therefore, tuning the conduction band energy level of the n-type buffer is critical for enabling efficient collection of photogenerated carriers. A “spike” structure is formed when the conduction band edge of the p-type absorber layer is lower than that of the n-type buffer layer ($E_{c,\text{absorber}} < E_{c,\text{buffer}}$) forming a barrier that impedes the collection of photogenerated carriers. Theoretical studies have predicted the optimal offsets to reside in the range between 0 eV and +0.4 eV.⁶ On the other hand, a “cliff” structure ($E_{c,\text{absorber}} > E_{c,\text{buffer}}$) increases charge recombination at the interface of a p-type absorber and n-type buffer.^{6,7}

The best performing SnS solar cells with device efficiencies of 4.63% were achieved using a p-SnS/n-SnO₂/n-Zn(O,S):N structure where Zn(O,S):N serves the role of the n-type buffer layer whose energy band levels can be tuned with varying compositions of O and S. Interfacial recombination can be suppressed by optimizing the heterojunction interface to have a small spike structure.^{8,9} The insertion

of an ultrathin SnO₂ atomic layer deposition (ALD) layer between the SnS and Zn(O,S):N layers was later discovered to significantly increase the open-circuit voltage (V_{OC}).¹⁰ It is believed that SnO₂ acts as a blocking layer that prevents the zinc diffusion into SnS during the Zn(O,S):N ALD process, resulting in increased recombination at the p-n interface and hence in the loss of the junction’s rectification properties. While a few monolayers of SnO₂ produced the highest device performance, the device suffered performance degradation when the oxide layer thickness is thick enough to form a conduction band. This can be explained by the unfavorable cliff structure of the SnS/SnO₂ junction disrupting the optimized spike structure formed at the SnS/Zn(O,S):N interface. Consequently, a small spike structure at the interface not only reduced charge recombination at the interface but also avoided impeding the efficient collection of the photo generated carriers with a sufficiently low barrier height at this interface.

Germanium-doped tin oxide is a logical advancement toward the design of an improved n-type buffer layer for SnS-based solar cell devices. This buffer layer material delivers two distinct advantages. First, there will be no potential diffusion of Zn like in the case of Zn(O,S):N. By replacing the Zn-based buffer layer with a tin-based material, we eliminate the problem of detrimental deep-level defects in SnS arising from zinc diffusing out of the buffer layer into the SnS absorber. Second, replacing the corrosive and toxic coreactant H₂S with a

noncorrosive organometallic ALD precursor for germanium enables greater control over the doping levels for finer CBO adjustments. Theoretical calculations also show the passivating effects that an oxide layer has on the surfaces of SnS where dangling bonds exist.¹¹ The optimal CBOs provided by an optimized Zn(O,S) layer are similarly achievable through fine-tuning the conduction band edges of SnO₂ by doping germanium into tin oxide, transforming the original cliff structure to a desired spike structure predominantly by raising the conduction band energy level. Therefore, to create the desired spike structure at the heterojunction interface, fine-tuning of composition is indispensable.

This work presents SnS-based photovoltaic devices using (Sn,Ge)O₂ as an n-type buffer layer. By leveraging the tunability of band alignments and adjustability of carrier densities, (Sn,Ge)O₂ is found to be a promising n-type buffer layer not only suitable for SnS but also highly applicable to various p-type absorber materials, such as Cu(In,Ga)(S,Se)₂ (CIGS)^{12,13} and Cu₂ZnSn(S₂,S₄)₄ (CZTS).^{14–16} ALD as the fabrication technique of choice for (Sn,Ge)O₂ provides precise control over the doping levels of Ge and N simply by adjusting the ratio of precursor pulses. The effects of germanium and nitrogen doping on the CBO and the conductivity of the materials were determined to guide this optimization.

An ALD process was used to deposit SnO₂ using tetrakis(dimethylamido)tin(IV) {[(CH₃)₂N]₄Sn, TDMASn, Sigma Aldrich} and hydrogen peroxide (H₂O₂, 30% w/w in H₂O, Sigma Aldrich). Vapors of TDMASn and H₂O₂ were injected sequentially into the deposition chamber. To prepare (Sn,Ge)O₂, germanium was doped into SnO₂ by injecting vapors of germanium(IV) methoxide [Ge(OCH₃)₄, Sigma Aldrich] and H₂O₂. TDMASn, Ge(OCH₃)₄, and H₂O₂ were kept at 60 °C, 30 °C, and 30 °C, respectively. Thermally oxidized silicon wafers, glassy carbon plates, and silica glass were utilized as substrates and were treated by a UV/ozone cleaner (wavelengths = 185 nm and 254 nm) for 5 min at room temperature before being introduced into the process chamber. The band alignments at the SnS/SnGeO₂ interface were determined by x-ray photoelectron spectroscopy (XPS, Thermo Scientific K-alpha XPS) films using bilayered films of SnS with (Sn,Ge)O₂ overlayers.¹⁷ The transmittance and reflectance spectra of 500-nm-thick SnS and 100-nm-thick (Sn,Ge)O₂ were measured at normal incidence using ultraviolet/visible and near-infrared spectrophotometry (Hitachi U-4100 UV-vis-NIR spectrophotometer) in the wavelength range of 200–800 nm. Film thicknesses were measured using x-ray reflectivity (PANalytical X-Pert Pro) and cross-sectional scanning electron microscopy (Carl Zeiss Ultra 55 FESEM).

A photovoltaic device structure of Si/SiO₂/Mo/SnS/(Sn,Ge)O₂/ZnO/ITO/Ag was used in this study. A Mo bilayer¹⁸ was first sputtered on silicon (100) substrates with 300 nm thick layers of thermally oxidized silicon dioxide on the surface. 500-nm-thick SnS thin film absorber layers were deposited at 120 °C using an ALD of bis(*N*, *N'*-diisopropylformamidinato)Sn(II) [Sn(famd)₂, Strem Chemicals] and hydrogen sulfide (H₂S, 4% in N₂) maintained at 60 °C and 25 °C, respectively. The SnS films were then annealed under a constant flow of 60 sccm of H₂S at 400 °C and 10 Torr pressure for an hour. The ZnO thin film was deposited at 120 °C by ALD using alternating doses of diethylzinc [Zn(C₂H₅)₂, DEZ, Sigma Aldrich] and de-ionized water vapor. 300 nm of tin-doped indium oxide (ITO) were deposited at room temperature via RF magnetron sputtering with a shadow mask to define an active device area of 0.24 cm². A 500-nm-thick Ag layer

acting as the top electrode was patterned by e-beam evaporation using another shadow mask at room temperature.

(Sn,Ge)O₂ thin films were prepared at a substrate temperature of 200 °C which is within the common temperature windows for the ALD processes of SnO₂ and GeO₂. Tin oxide films have been deposited by ALD at the process temperatures of 50–300 °C using TDMA(Sn) as the tin source.¹⁹ Ge(OCH₃)₄ was discovered to not deposit a GeO₂ film on nonhydroxylated surfaces at substrate temperatures of 200–300 °C. Instead, the precursor required the OH-terminated surfaces of ALD-SnO₂ and temperatures of at least 200 °C for the reaction with water vapor to occur and promote the incorporation of germanium into SnO₂. The lower end of the overlapping ALD temperature window of 200 °C was selected to exploit the reduced doping efficiency of germanium, allowing us to establish finer control over film compositions and access more uniform distribution of dopants throughout the film.

Figure S1 shows that the germanium content in the (Sn,Ge)O₂ film is approximately equal to the precursor pulse ratio, indicating excellent doping efficiency of Ge in the presence of OH-terminated surfaces during the Ge precursor pulse, and slightly more so in films with a greater amount of Ge incorporated. Carbon and nitrogen peaks were observed only on the surface of the films due to exposure to ambient contaminants [Figs. S2(a) and S2(b)]. The absence of these contaminants in the bulk indicates the high purity of the (Sn,Ge)O₂ films as a result of the clean and complete reaction between both tin and germanium precursors with H₂O₂ when operating within the common temperature windows of both ALD processes.

The x-ray diffraction pattern of (Sn,Ge)O₂ films deposited on 1 × 1 in.² quartz substrates exhibits no specific peaks from the film, which suggests a nearly amorphous or nanocrystalline structure of the film.

The carrier mobility, electron carrier density, and resistivity are plotted as a function of the Ge/(Ge + Sn) ratio in Fig. 1(a). Hall measurements demonstrate that the (Sn,Ge)O₂ films have n-type conductivities with electron concentrations tunable between 10¹⁸ cm⁻³ and 10²⁰ cm⁻³. The electron Hall mobility varied only slightly with varying amounts of germanium incorporation in the film which is expected as the material remains amorphous independent of germanium doping levels. The germanium doped tin oxide films showed a lower carrier concentration compared to the undoped tin oxide films. Since both tin and germanium cations are in the same +4 oxidation state, we do not expect any increase in the electron concentration from germanium doping. Instead, the conductivity of SnO₂ films is attributed to the presence of oxygen vacancies in the material.²⁰

Deconvolution of the O-1s XPS spectra of (Sn,Ge)O₂ ALD films as presented in Figs. S3(a)–S3(c) shows two components with peak centers at 531.2 eV and 532.7 eV, respectively. Neither of these photoemission lines can be assigned to O-Ge bonds as both lines exist in the O-1s region of undoped SnO₂, with the higher energy emission reducing in intensity as more Ge is doped into the material, in contrast to the expected behavior of the O-Ge line. Instead, the lower energy peak O_L is assigned to the oxygen ions bonded to Sn ions, while the higher energy peak O_{OV} can be attributed to the presence of oxygen vacancies in the SnO₂ sublattice. As the Ge content of the film is increased from 0 to 14 at. %, the O_{OV} peak assigned to oxygen vacancies reduced from 59% in the undoped SnO₂ film to 2% in the film with 14 at. % of germanium, corresponding to a significant reduction in the concentration

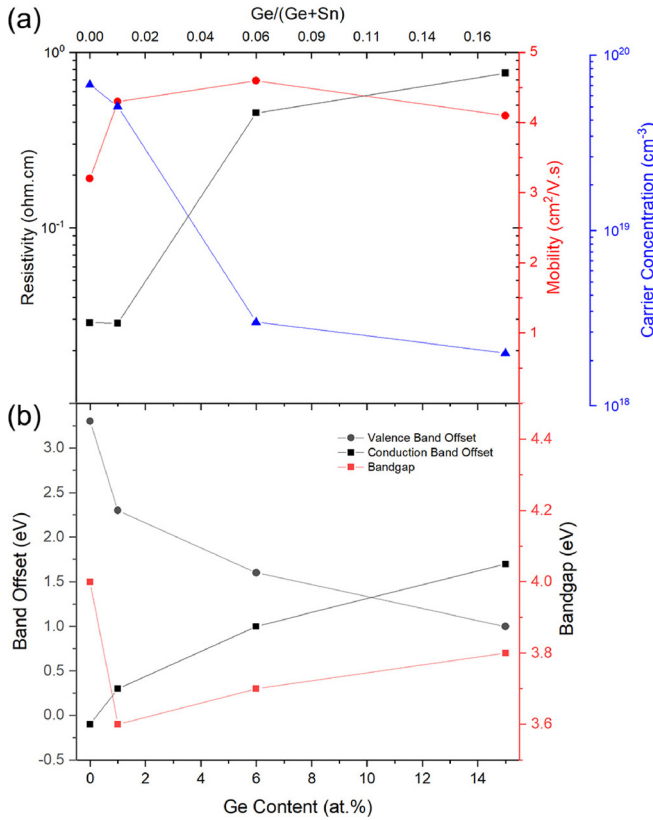


FIG. 1. Effect of various amounts of germanium incorporation in SnO₂ films on (a) Hall mobility, carrier concentration, and resistivity and (b) optical bandgaps as determined from absorption data and band offsets of both the valence bands and conduction bands at the heterojunction formed with SnS.

of oxygen vacancies. This is explained by the bond strength of Ge-O (657.5 kJ mol⁻¹) which is greater than that of Sn-O (528 kJ mol⁻¹), making it more effective at capturing oxygen and hence suppressing the formation of oxygen vacancies. The electrical conductivity of SnO₂ is typically attributed to the formation of oxygen vacancies, given by the equation

$$\text{O}_\text{O}^\times = \frac{1}{2} \text{O}_2(\text{g}) + \text{V}_\text{O}^{**} + 2\text{e}^-,$$

where two electron carriers and a doubly charged oxygen vacancy are the result of the loss into the gas phase of an oxygen unit for each SnO₂ unit in the material. Thus, when the germanium doping is increased, the resulting reduction in oxygen vacancies corresponds to a decrease in the electron carrier concentration of the film, giving rise to an increased electrical resistivity.

The absorption coefficient α is obtained from the optical transmission and reflectance measurements and calculated using

$$\alpha = \frac{1}{d} \ln \left(\frac{1-R}{T} \right),$$

where d is the film thickness and R and T are the reflectance and transmittance of the film, respectively. The optical bandgap energy value is

obtained from plots of α^2 vs photon energy for the various ALD films deposited on quartz substrates, using the Tauc relation for direct transitions²¹

$$\alpha(h\nu) \propto (h\nu - E_g)^{\frac{1}{2}},$$

where $\alpha(h\nu)$ is the absorption coefficient, $h\nu$ is the photon energy, and E_g is the optical bandgap. The bandgap is then obtained by extrapolating to $\alpha \cdot h\nu = 0$. Bilayer samples of 500-nm-thick SnS and 5-nm-thick SnO₂ with various concentrations of germanium dopants were fabricated to investigate the electronic band alignment of (Sn,Ge)O₂ relative to SnS. The valence band offset (ΔE_{VB}) and conduction band offset (ΔE_{CB}) in the SnS/(Sn,Ge)O₂ stack are, respectively, calculated using the relations

$$\Delta E_{VB} = \left(E_{\text{SnS}/(\text{Sn,Ge})\text{O}_2}^{\text{SnS}} - E_{\text{Ge-3d}}^{\text{SnS}/(\text{Sn,Ge})\text{O}_2} \right) + \left(E_{\text{Ge-3d}}^{(\text{Sn,Ge})\text{O}_2} - E_{\text{VBM}}^{(\text{Sn,Ge})\text{O}_2} \right) - \left(E_{\text{Sn-3d}}^{\text{SnS}} - E_{\text{VBM}}^{\text{SnS}} \right),$$

$$\Delta E_{CB} = \left(E_{\text{SnS}/(\text{Sn,Ge})\text{O}_2}^{\text{SnS}} - E_{\text{Ge-3d}}^{\text{SnS}/(\text{Sn,Ge})\text{O}_2} \right) + \left(E_{\text{Ge-3d}}^{(\text{Sn,Ge})\text{O}_2} - E_{\text{VBM}}^{(\text{Sn,Ge})\text{O}_2} \right) - \left(E_{\text{Sn-3d}}^{\text{SnS}} - E_{\text{VBM}}^{\text{SnS}} \right) + \left(E_g^{(\text{Sn,Ge})\text{O}_2} - E_g^{\text{SnS}} \right),$$

where $E_g^{(\text{Sn,Ge})\text{O}_2}$ and E_g^{SnS} are the optical bandgaps of bulk (Sn,Ge)O₂ and SnS, respectively.²²

Bandgap bowing is demonstrated in (Sn,Ge)O₂ in Fig. 1(b) where the bandgap of SnO₂ narrows initially when germanium is added to SnO₂ before widening toward that of GeO₂ [5.8 eV (Ref. 23)] to complete the bowing. When increasing the doping levels of germanium reduces the electron concentrations to under 10¹⁹ cm⁻³, the decreased influence of the Burstein-Moss effect is believed to be competing with the bandgap widening, contributing to the bowing effect of the bandgap. For all values of the germanium content studied, the bandgap of (Sn,Ge)O₂ remains above 3.5 eV, making SnO₂ in the entire doping range highly suitable as a window layer of solar cells by possessing excellent transparency in the AM 1.5 spectrum. Figure 1(b) shows that (Sn,Ge)O₂ films exhibit a monotonic decrease in the valence band offsets relative to SnS, resulting in an overall monotonic increase in CBOs with the increasing Ge content of the SnO₂ film. An almost-linear relationship is observed between the CBOs and the amount of germanium doping despite bandgap bowing. This makes it easy to tune the CBO simply by controlling the SnO₂/GeO₂ ALD cycle ratio.

The optimal composition of the (Sn,Ge)O₂ buffer layer for partnering with SnS was determined by tuning the energy band alignments at the SnS/(Sn,Ge)O₂ heterojunction. Devices with an active area of 0.24 cm² (Fig. 2) were fabricated using ALD SnS absorbers with different elemental compositions of (Sn,Ge)O₂ buffer layers. SnO₂:GeO₂ ALD subcycle ratios of 59:1, 39:1, 19:1, 9:1, 3:1, and 1:1 were used to fabricate the devices, and a germanium-free SnO₂ buffer layer is used in the baseline device. The amount of germanium in these films for the various SnO₂:GeO₂ ALD subcycle ratios can be found in Fig. S1. The undoped SnO₂ layer formed a type-II cliff structure with SnS due to having a slightly negative CBO, causing the device to suffer from increased interfacial recombination near the p-n junction. While SnO₂ provides beneficial passivating effects at the interface between SnS and Zn(O,S), it lacked the appropriate band alignment with SnS and resulted in nonrectifying devices when the SnO₂ layer exceeded several

nanometers in film thickness. Since germanium has demonstrated the ability to raise the conduction band energy level of SnO_2 , doping germanium into the film shifts the heterojunction into a type-I scenario with a positive CBO, improving the rectification and hence the performance of the device. However, as we decrease the $\text{SnO}_2/\text{GeO}_2$ ALD subcycle ratio and consequently incorporate too much germanium into the material, the CBO with SnS increases considerably to the point where photo generated electrons are being impeded. Although having a $(\text{Sn,Ge})\text{O}_2$ buffer layer with an ALD cycle ratio of 19:1 demonstrated modest J-V characteristics under illumination relative to the undoped SnO_2 layer, the device had a low fill factor of 25% and displayed a lack-luster power conversion efficiency (PCE) of 0.4%. Decreasing the amount of germanium content by switching to a $\text{SnO}_2/\text{GeO}_2$ ALD cycle ratio of 29:1 is expected to provide improved performances with the presence of a sufficiently small conduction band spike at the heterojunction, but the actual device exhibited a poor ohmic J-V response. With a $\text{SnO}_2/\text{GeO}_2$ cycle ratio of 29:1, the electron density of $(\text{Sn,Ge})\text{O}_2$ is higher than that of the film with a 19:1 cycle ratio. Doping $(\text{Sn,Ge})\text{O}_2$ with nitrogen provided a significant recovery of the diode quality by decreasing the conductivity of the buffer layer. Nitrogen is introduced into the lattice of $(\text{Sn,Ge})\text{O}_2$ by injecting an additional dose of purified NH_3 gas immediately following the water vapor dose at the end of each GeO_2 ALD subcycle. Nitrogen incorporation in the film serves to reduce the large electron density from $5 \times 10^{19} \text{ cm}^{-3}$ down an order of magnitude to $6 \times 10^{18} \text{ cm}^{-3}$ to compensate for the higher carrier density due to the reduction in germanium doping. Charge compensation occurs when acceptor levels that are created in the SnO_2 bandgap neutralize the electron density due to oxygen vacancies, leading to the reduced carrier density in the material.²⁴

The device utilizing the nitrogen doped $(\text{Sn,Ge})\text{O}_2$ buffer layer with an ALD subcycle ratio of 29:1 demonstrates the best photovoltaic performance with $J_{\text{sc}} = 12.2 \text{ mA/cm}^2$, $V_{\text{oc}} = 0.40 \text{ V}$, $\text{FF} = 46\%$, and $\text{PCE} = 2.21\%$. This device has the highest reported V_{oc} to date for a SnS-based solar cell. The JV-characteristics in Fig. 2 demonstrate how tuning both the carrier concentration and band energy alignments at the heterojunction is vital to optimizing the photovoltaic device efficiencies and achieving high V_{oc} performance.

In conclusion, we have demonstrated the atomic layer deposition of $(\text{Sn,Ge})\text{O}_2$ as a buffer layer for SnS-based solar cells. Significant improvement in V_{oc} can be captured through optimization of the conduction band alignments and carrier concentration of the buffer layer, achieving a V_{oc} of 0.40 V and a device efficiency of 2.21%. Germanium doping raises the conduction band level of SnO_2 while reducing the electron concentration by reducing the oxygen vacancies in the film. By incorporating nitrogen into the film using ammonia gas, the carrier concentration of $(\text{Sn,Ge})\text{O}_2$ can be reduced to match the absorber layer. Our results highlight the importance and advantages of exploiting an electrically and energetically tunable buffer material such as $(\text{Sn,Ge})\text{O}_2$ with an adjustable conduction band energy level, enabling the CBO to be adapted to a wide range of absorber materials and surfaces, while at the same time being able to change the carrier concentration to match that of the absorber material. Furthermore, eliminating the use of zinc in the n-type partner layer avoids midgap defects associated with Zn diffusing out from $\text{Zn}(\text{O,S})$ buffer layers across the p-n junction. By simultaneously serving as a passivating layer, a diffusion barrier, and an electron transport layer with tunable band alignments and adjustable carrier concentrations to

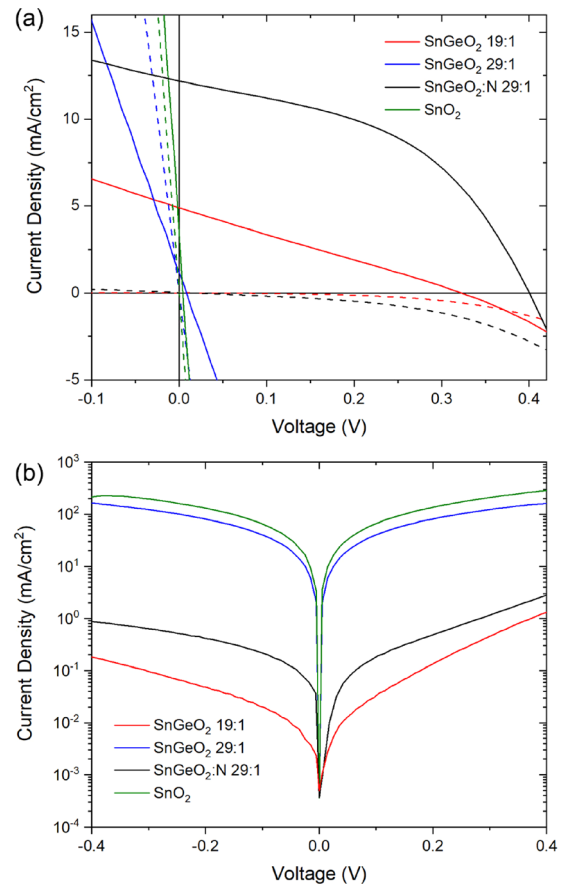


FIG. 2. Current density-voltage characteristics of Mo (500 nm)/SnS (500 nm)/(Sn,Ge) O_2 (40 nm)/ZnO (10 nm)/ITO (200 nm)/Ag (500 nm) devices at various (Sn,Ge) O_2 compositions (a) under dark (dotted line) and AM 1.5 illumination (solid line) on a linear scale and (b) under dark conditions on a semilog scale.

match SnS, $(\text{Sn,Ge})\text{O}_2$ has demonstrated potential as a partner material for SnS absorber layers in solar cells.

See the [supplementary material](#) for details on the influence of ALD subcycle ratios on the film composition, XPS studies on the chemical purity of the film and on the effect of Ge doping on the hydroxyl content, cross-sectional SEM study of the solar cell device stack, optical properties of the $(\text{Sn,Ge})\text{O}_2$ film, and optimization of the $(\text{Sn,Ge})\text{O}_2$ layer thickness in the SnS-based photovoltaic devices.

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